Evaluation report — BTCUSDT X-sentiment price model

CrowdWisdomTrading · X Sentiment Price Prediction assessment
Asset BTCUSDT · 1-hour bars · UTC · horizons n+0 … n+5
Repository: https://github.com/AtelierGivre/BTC-X-Sentiment-Price-Prediction-Technical-Build-Reference
Generated 2026-11-06 · 25 tests passing

Summary

- **Does sentiment add value?** Against the identical model on price only, Ridge wins in 0/6 horizons and LightGBM in 2/6 (lower RMSE ratio).
- **Best forecast:** `ridge_price_only` at h=0 has RMSE ratio 1.001 vs naive (1.000 = no improvement) and directional accuracy 48.2%.
- **Best tradable strategy:** `ridge_price_only__long_flat` — Sharpe -8.02 (95% CI -13.37 to -2.32), Sortino -10.27, Max DD -17.9%, net -16.7% vs -10.7% gross (fees 6.9%).
- **Buy-and-hold:** Sharpe -4.40, Max DD -11.6%, net -10.4%.

A gross edge smaller than the fees needed to harvest it cannot be traded profitably. The bootstrap interval, not the point estimate, is the decision rule; with a short sample a wide interval spanning zero is the expected outcome and is reported as such.

1. Objective and setup

Predict BTC's price at **n+0 … n+5 hours** from **X sentiment and price only**. Bars are 1 hour, labelled by open time in UTC. At the start of bar `n` only information up to the close of bar `n-1` is used. Target: `y_h = log(close[n+h] / close[n-1])` for h = [0, 1, 2, 3, 4, 5]. Costs are 7 bps per position change.

2. Data

Price source: Binance BTCUSDT hourly klines (fallback yfinance BTC-USD). Full history: 180 days (~4,320 bars, free); the model uses the tweet window plus 24h warm-up (~60 days), so no hour is trained on a fabricated zero-sentiment bar.

Tweet source: Apify X scrape, actor `xquik~x-tweet-scraper` (~$0.15/1K tweets). Actual coverage: **28,342 tweets over 60.0 days** (2026-09-06 → 2026-11-05 UTC), cost **$4.25**. A 180-day scrape was not affordable on the Apify FREE plan ($5/month), so the model is trained only on the tweet window (plus 24h of price warm-up); the rest of the price history is unused. This is disclosed, not silently back-filled with another dataset. Sentiment: VADER.

Row counts through cleaning:

step	rows
start	28,342
after_dedup_tweet_id	28,342
after_nonempty	28,331
after_english_filter	28,312
after_author_repeats	28,291
after_swarm_dedup	28,291
after_spam_filter	28,291

3. Sentiment method

VADER compound score in [-1, 1] per tweet. Hourly aggregates: tweet count, mean/median/std, positive/negative share, follower-weighted and engagement-weighted mean. Optional transformer scores are compared with VADER when `TRANSFORMER_MODEL` is set.

4. Features and leakage

56 features: price returns at lags [1, 2, 3, 6, 12, 24], 6/24h volatility, 24h momentum, volume z-score; the same lags and moving averages for four sentiment series (`sent_mean`, `w_follow`, `w_engage`, `pct_pos`) plus a 168h z-score and 1h change; tweet volume (log and z-score); a no-tweets flag; hour-of-day and weekday. **All feature columns receive one global `.shift(1)`**, and calendar features are added after the shift. `tests/test_leakage.py` corrupts all data from bar `n` onwards and asserts features at `n` are unchanged.

5. Validation design

Expanding walk-forward: first train about 30 days (adaptive: 90 days capped at half the sample), test 7 days, step 7 days, **6-bar purge gap**. 4 folds:

fold	test_start	test_end
0	2026-10-06 10:00:00	2026-10-13 09:00:00
1	2026-10-13 10:00:00	2026-10-20 09:00:00
2	2026-10-20 10:00:00	2026-10-27 09:00:00
3	2026-10-27 10:00:00	2026-11-03 09:00:00

Scalers and RidgeCV penalties are fitted inside each training window only.

6. Results — prediction (out-of-sample)

RMSE ratio vs the naive (zero-return) benchmark by horizon and model:

h	naive	ridge	ridge_price_only	lgbm	lgbm_price_only
0	1.000	1.001	1.001	1.028	1.031
1	1.000	1.002	1.002	1.053	1.047
2	1.000	1.003	1.003	1.063	1.064
3	1.000	1.004	1.004	1.083	1.072
4	1.000	1.005	1.004	1.092	1.071
5	1.000	1.006	1.005	1.090	1.071

A ratio at or above 1.000 means the model is no better than predicting a zero return; the best model here reaches 1.001, so no forecast beats the naive benchmark at any horizon. Full metrics (RMSE, MAE, directional accuracy, Spearman IC, USD MAE) are in `reports/metrics_prediction.csv`.

7. Results — strategy (after costs)

`sharpe` is the **Sharpe Ratio**, `sortino` is the **Sortino Ratio**, and `max_drawdown` is the **Max Drawdown** (peak-to-trough on the compounded equity curve). All are annualised from hourly net returns.

strategy	sharpe	sharpe_ci_low	sharpe_ci_high	sortino	max_drawdown	total_return	gross_return	fee_drag	n_trades	n_hours
buy_hold	-4.401	-9.298	1.900	-5.816	-0.116	-0.104	-0.104	0.000	1	672
ridge__long_short	-21.331	-29.255	-14.068	-25.026	-0.424	-0.416	-0.085	0.449	323	672
ridge__long_flat	-15.180	-21.615	-8.653	-18.676	-0.287	-0.277	-0.094	0.225	322	672
lgbm__long_short	-14.569	-19.176	-9.135	-18.621	-0.314	-0.309	0.041	0.409	294	672
lgbm__long_flat	-12.646	-17.807	-7.362	-16.179	-0.221	-0.212	-0.034	0.204	291	672
ridge_price_only__long_short	-10.102	-15.608	-5.079	-12.689	-0.238	-0.224	-0.111	0.136	99	672
ridge_price_only__long_flat	-8.020	-13.369	-2.323	-10.267	-0.179	-0.167	-0.107	0.069	99	672
lgbm_price_only__long_short	-16.657	-22.295	-10.464	-20.872	-0.354	-0.347	-0.026	0.400	288	672
lgbm_price_only__long_flat	-14.136	-18.808	-8.826	-17.728	-0.239	-0.235	-0.066	0.200	286	672

`fee_drag` is total fees in log-return units (7 bps × turnover); `gross_return` adds it back, so the gap between gross and net is the cost of trading. All positions use the h=0 forecast (the return of the current bar), entered at the start of the bar. Bootstrap intervals are 95% moving-block bootstrap (24h blocks, 1,000 resamples) on the Sharpe.

8. Ablations

`ridge_price_only` and `lgbm_price_only` use the same models without any sentiment column. Comparing them with `ridge` / `lgbm` answers the real question: **does sentiment add anything beyond price?** Read the RMSE-ratio and Sharpe differences, not just the best absolute number. LightGBM attributes 58% of its split gain to sentiment features, but the price-only ablation matches or beats the full model — gain share measures how often a feature is used, not whether it helps out of sample.

9. Charts

Interactive charts are in `reports/figures/`: 1. sentiment vs price, 2. tweet volume, 3. rolling sentiment/return correlation, 4. predicted vs actual, 5. equity curves + drawdown, 6. metric matrix, 7. feature importance, 8. sentiment-change vs price information matrix (sentiment lag × forecast horizon).

10. Limitations

The tweet sample covers 60 days, not 180, because of the Apify FREE credit; price history is longer but sentiment only exists inside that window. The scrape is a `Top`-ranked sample (engagement-biased) capped at ~480 tweets/day, so quiet hours are thin. Other limits: one market regime, bot noise, X search coverage, fees assumed at 7 bps with no slippage, and fold-level position resets. A weak or null result is acceptable if the method is clean.

11. Reproduction

```
python run_pipeline.py --stage ingest
python run_pipeline.py --stage process
python run_pipeline.py --stage train
python run_pipeline.py --stage report
pytest -q
```

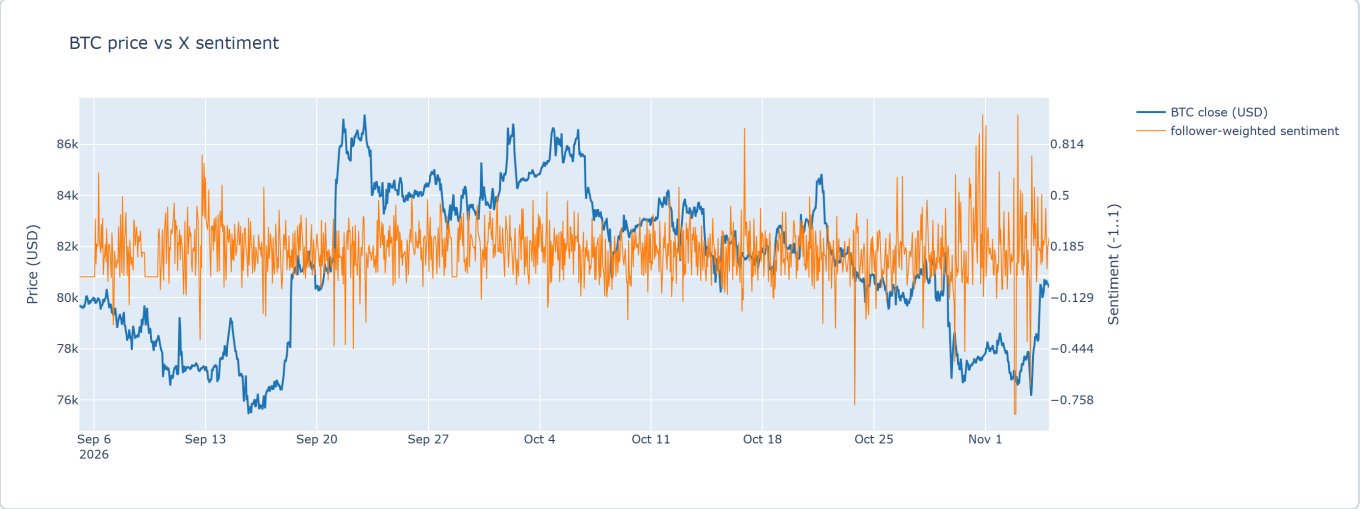
Small pre-computed tables (`master`, `features`, `predictions`, `strategy_returns`, `feature_importance`) are committed under `artifacts/`, so the report can be reproduced without re-scraping X and without spending Apify credit.

12. Latest forecast

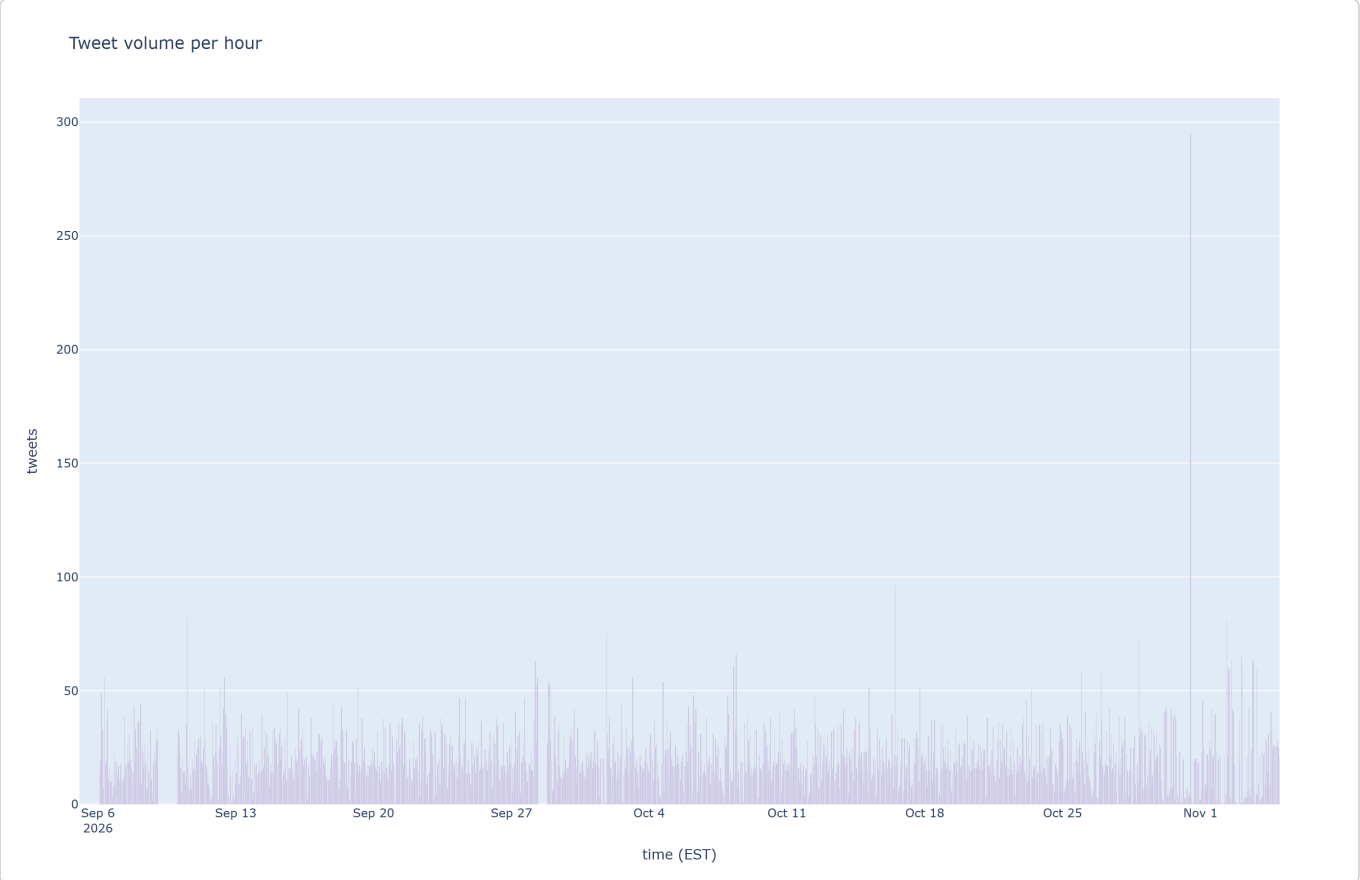
bar	h	ridge_ret	ridge_price	lgbm_ret	lgbm_price
2026-11-05 06:00:00+00:00	0	0.000094	80407.539294	0.000736	80459.209331
2026-11-05 06:00:00+00:00	1	-0.000074	80394.046471	0.001395	80512.269142
2026-11-05 06:00:00+00:00	2	-0.000127	80389.801139	0.000969	80477.980326
2026-11-05 06:00:00+00:00	3	-0.000009	80399.301677	0.001203	80496.797381
2026-11-05 06:00:00+00:00	4	0.000181	80414.561472	0.003505	80682.321035
2026-11-05 06:00:00+00:00	5	0.000383	80430.801789	0.003069	80647.153394

Appendix — charts

Static exports of the eight interactive charts in `reports/figures/`.



01_sentiment_vs_price



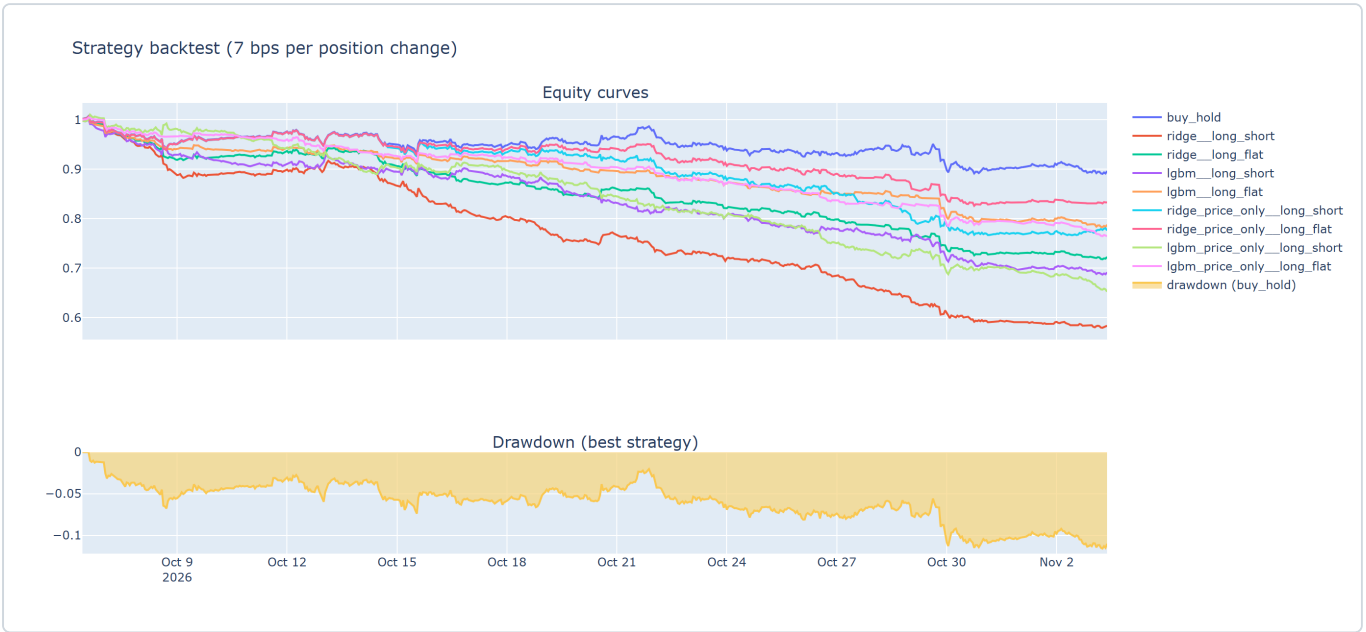
02_tweet_volume



03_rolling_correlation



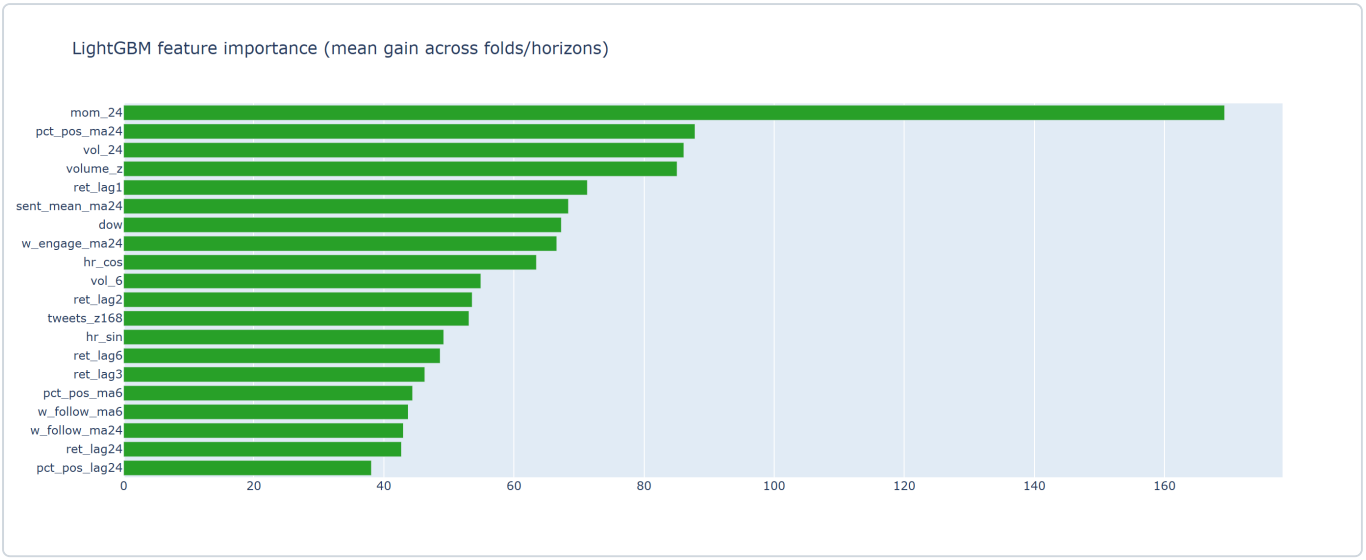
04_pred_vs_actual



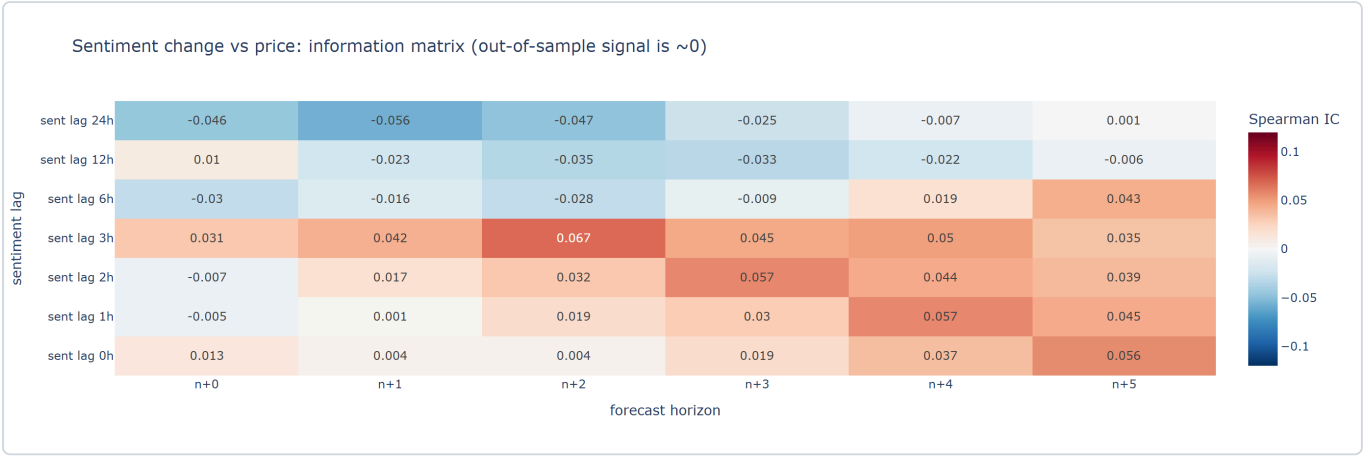
05_equity_curves



06_metric_matrix



07_feature_importance



08_sentiment_change_matrix